

5.9 TONS of carbon produced per person, per year in the US.

2.9 TONS of carbon produced per person, per year in the UK.

0.3 TONS of carbon produced per person, per year in India.

world's forests are lost each year to industrial logging, so less and less carbon is being neutralized.

The oceans also absorb CO₂ and contain the largest store of the gas on Earth, but warming of the oceans reduces the sea's capacity to absorb carbon dioxide from the atmosphere. There has been widespread retreat of glaciers in nonpolar regions, and a

of millions of people from low-lying areas such as the Ganges and Nile deltas.

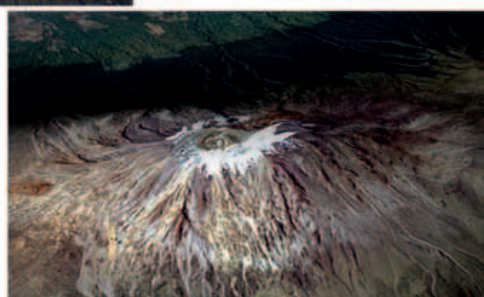
Many animal species are in danger of extinction. The World Conservation Union (WCU) considers 688 African fish, bird, and mammal species to be under threat, and 201 of these are classed as "critically endangered." The reduction of Arctic summer sea ice is jeopardizing the survival of some species, including polar bears. The IPCC suggests that the summer sea ice will disappear entirely in the second half of the 21st century. The message from environmental lobby groups worldwide is that governments must act, and act now, to reduce CO₂ emissions.

Industrialized countries are promoting personal carbon-offset programs in which people are encouraged to plant trees to offset their "carbon footprints." However, given the number of trees needed to offset the carbon emissions of just one car per year and fact that large



The Shining Mountain

According to some scientists, Mount Kilimanjaro's ice fields could be gone by the year 2020. The ice on the mountain's summit has dwindled by 82 percent over the past century, which is particularly remarkable given that it has survived previous climate shifts—including a severe 300-year-long drought 4,000 years ago. These images show the peak in February 1993 (top) and February 2000 (right).



decrease in global snow and ice cover since the late 1960s. Sea levels are rising, and may rise a further 20–16 in (25–40 cm) by 2100.

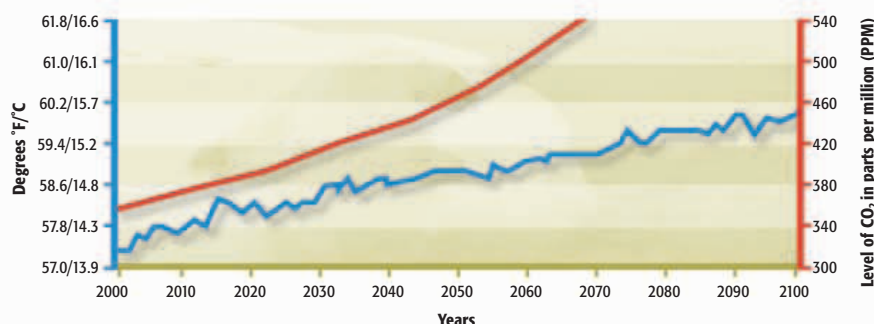
Experts also believe climate change is responsible for changes in rainfall patterns, which have created severe water shortages in many regions of the world—tens of millions have suffered from drought in Africa since the 1980s. Climate change probably contributes to the increased intensity of heatwaves and tropical storms, although no direct causal link has been established with such extreme weather events as Hurricane Katrina, which devastated the north-central Gulf Coast in 2006.

The implications

The IPCC and many other organizations predict dire consequences for the world, and especially for the poorest nations, if CO₂ emissions continue to rise. Increasing sea levels could displace tens

Lengthy fast

An adult female polar bear hauls herself out of the Arctic waters. The progressively earlier breakup of sea ice shortens the vital spring hunting season for female polar bears. As a result, weight loss is affecting their ability to reproduce, and the survival of their cubs.



Years

KEY

— CO₂ concentration

— Temperature

21st century warming

Current projections for global warming predict temperature rises of between 3.2 and 7°F (1.8–4°C) and a steady rise in atmospheric concentrations of CO₂. The increased levels of this gas in the atmosphere heighten the greenhouse effect, pushing temperatures upward.

portions of the world's forests are disappearing, the fear is that climate change may now be progressing so rapidly that humans will soon be unable to stop it.

The scientific debate

The factors that affect climate are complex, and the scientific community is not universally persuaded that human beings are—or have been—a major cause of climate change. Natural climate variability is certainly a factor, but most scientists believe that Earth is undergoing significant warming as a direct result of human activities—and that this trend of global warming will continue.

32 MILLION acres of forest lost each year.

240 TREES needed to absorb the CO₂ emitted by one car each year.

10 PERCENT of the world's coral reefs now destroyed.

“Future generations may well have occasion to ask themselves, ‘What were our parents thinking?’ ”

AL GORE, FROM “AN INCONVENIENT TRUTH”, 2006

HOW WE KNOW

MEASURING CLIMATE CHANGE

Scientists monitor glaciers in the Arctic and Antarctica to determine the current changes occurring in the polar regions. However, in order to assess climate change, they must form a picture of past climates. Clues to past climatic conditions are hidden in rocks and trees, but the most revealing indicators of the historical climate are preserved in ice (see pp.22–23). Scientists studying glaciers are able to analyze air bubbles trapped in the ice and determine the levels of gases in the atmosphere at the time the ice was formed, giving them information on how the atmosphere has changed over the centuries.



AFTER

Given humankind's destructive impact on the environment, the race is on to find alternative sources of energy to replace power generation using fossil fuels.

TIDAL POWER

Investigations into harnessing tidal stream power, where energy is drawn from tidal currents, are showing promise, and prototype **wave energy converters** are also producing good results, but both these techniques are at an early stage of development. At La Rance, France, a tidal barrier built in the 1960s generates 240MW of power, but such barrages are expensive to build.

SOLAR POWER

Photovoltaic cells convert solar energy into electricity and provide a **clean source of electricity**, but the technology is expensive and cannot provide enough power for large commercial needs. In principle, however, much more solar energy falls on Earth than the amount of energy generated by burning fossil fuels; so solar power generation remains a key area of research.

WIND POWER

The conversion of wind energy into electricity currently provides just one percent of the electricity used worldwide. However, it is a popular form of **sustainable energy** in certain European nations. Nine percent of the electricity used in Spain and 20 percent of electricity used in Denmark is generated by wind power. This technology is not without drawbacks—the construction of the platforms for each turbine uses large amounts of cement, a major contributor to CO₂ emissions.



WIND TURBINES